

EXPERIMENT-1

.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <title>Responsive Web Page</title>
</head>
<body style="background-color:#f2f2f2 ; ">
  <div class="container border border-dark mt-2">
    <div style="background-color:#2f80ed;color:white ; text-align:center ;
padding:40px ; ">
      <h1>My Website</h1>
    </div>
    <nav class="navbar navbar-expand-lg" style="background-color:#222 ; ">
      <div class="container-fluid">
        <a class="navbar-brand text-white" href="#">Website</a>
        <button class="navbar-toggler bg-light" type="button" data-bs-
toggle="collapse" data-bs-target="#navbarNav">
          <span class="navbar-toggler-icon"></span>
        </button>
        <div class="collapse navbar-collapse justify-content-end"
id="navbarNav">
          <ul class="navbar-nav">
```

```

<li class="nav-item">
    <a class="nav-link text-white" href="#">Home</a>
</li>
<li class="nav-item">
    <a class="nav-link text-white" href="#">About</a>
</li>
<li class="nav-item">
    <a class="nav-link text-white" href="#">Services</a>
</li>
<li class="nav-item">
    <a class="nav-link text-white" href="#">Contact</a>
</li>
</ul>
</div>
</div>
</nav>

<div style="padding:40px 20px ; background-color:#e9e9e9 ; min-
height:250px ; ">
    <div class="row">
        <div class="col-sm-12 col-md-6" style="padding:20px ; border-right:2px
solid gray ; ">
            <h4>Column 1</h4>
            <p>
                This is the first column. On small devices, it will take full width,
                while on medium and larger devices, it occupies half width using Bootstrap grid
                classes.
            </p>
        </div>
        <div class="col-sm-12 col-md-6" style="padding:20px ; ">
            <h4>Column 2</h4>

```

<p>

This is the second column. Bootstrap grid system helps to create responsive layout.

</p>

</div>

</div>

</div>

<div class="Footer">

<p class="text-center m-0 p-3" style="background-color:#222 ; color:white ; ">

© 2026 My Website.

</p>

</div>

</div>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
></script>

</body>

EXPERIMENT-2

.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <title>TechStore</title>
</head>
<body style="background-color:#f4f4f4;">
  <nav class="navbar navbar-expand-lg"
    style="background-color:#1f2329;">
    <div class="container">
      <a class="navbar-brand text-white" href="#">TechStore</a>
      <button class="navbar-toggler bg-light" type="button" data-bs-
toggle="collapse" data-bs-target="#navbarNav">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="collapse navbar-collapse justify-content-end"
        id="navbarNav">
        <ul class="navbar-nav">
          <li class="nav-item">
            <a class="nav-link text-white" href="#">Home</a>
          </li>
```

```
<li class="nav-item">
  <a class="nav-link text-white" href="#">Products</a>
</li>
<li class="nav-item">
  <a class="nav-link text-white" href="#">About</a>
</li>
<li class="nav-item">
  <a class="nav-link text-white" href="#">Contact</a>
</li>
</ul>
</div>
</div>
</nav>
<div class="container py-5">
  <div class="row g-4">
    <div class="col-sm-12 col-md-6 col-lg-4">
      <div class="card h-100">
        
        <div class="card-body">
          <h4>Laptop</h4>
          <p>High performance laptop</p>
          <h5>
            ₹50,000
            <span class="badge bg-success">New</span>
          </h5>
          <button class="btn btn-primary mt-2"
            data-bs-toggle="modal"
```

data-bs-target="#productModal">

View Details

</button>

</div>

</div>

</div>

<div class="col-sm-12 col-md-6 col-lg-4">

<div class="card h-100">

<div class="card-body">

<h4>Mobile</h4>

<p>Latest smartphone</p>

<h5>

₹30,000

Sale

</h5>

<button class="btn btn-primary mt-2"

data-bs-toggle="modal"

data-bs-target="#productModal">

View Details

</button>

</div>

</div>

</div>

<div class="col-sm-12 col-md-6 col-lg-4">

<div class="card h-100">

![Headphones](https://images.unsplash.com/photo-1505740420928-5e560c06d30e)

<div class="card-body">

<h4>Headphones</h4>

<p>Noise cancelling</p>

<h5>

₹5,000

New

</h5>

<button class="btn btn-primary mt-2"

data-bs-toggle="modal"

data-bs-target="#productModal">

View Details

</button>

</div>

</div>

</div>

</div>

</div>

<div class="modal fade" id="productModal">

<div class="modal-dialog">

<div class="modal-content">

<div class="modal-header">

<h5 class="modal-title">

Product Details

</h5>

<button type="button"

class="btn-close"

```
        data-bs-dismiss="modal">
    </button>
</div>
<div class="modal-body">
    This product is available with premium quality and latest features.
</div>
<div class="modal-footer">
    <button class="btn btn-secondary"
        data-bs-dismiss="modal">
        Close
    </button>
</div>
</div>
</div>
</div>
<div style="background-color:#1f2329;
    color:white;
    text-align:center;
    padding:15px;">

    © 2026 TechStore | Follow us on social media

</div>
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
></script>
</body>
</html>
```


EXPERIMENT-3

.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <script src="https://cdn.tailwindcss.com"></script>
  <title>FitLife</title>
</head>
<body style="background-color:#f3f4f6;">
  <header style="background-color:#1f2937;
    color:white;">
    <div class="container mx-auto flex flex-col md:flex-row justify-between items-
center p-4">
      <h1 style="font-size:32px;
        font-weight:bold;">
        FitLife
      </h1>
      <nav class="mt-3 md:mt-0">
        <ul class="flex space-x-4">
          <li><a href="#" style="color:white;">Home</a></li>
          <li><a href="#" style="color:white;">Programs</a></li>
          <li><a href="#" style="color:white;">Trainers</a></li>
          <li><a href="#" style="color:white;">Contact</a></li>
        </ul>
```

</nav>

</div>

</header>

<section style="background:linear-gradient(to right,#bfdbfe,#818cf8);text-align:center;padding:80px 20px;">

<h2 style="font-size:48px;font-weight:bold;color:white;">Transform Your Body & Mind</h2>

<p style="color:white;
margin-top:15px;
font-size:18px;">

Join FitLife today and achieve your fitness goals with expert trainers.

</p>

<button style="margin-top:25px;
background-color:white;
color:#1f2937;
padding:12px 24px;
border:none;
border-radius:8px;
cursor:pointer;">

Join Now

</button>

</section>

<!-- Programs -->

<section style="padding:60px 20px;">

<h2 style="text-align:center;
font-size:36px;
font-weight:bold;
margin-bottom:40px;">

Our Programs

</h2>

<div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">

<!-- Card 1 -->

<div style="background-color:white;

padding:25px;

border-radius:10px;

text-align:center;

box-shadow:0 0 10px rgba(0,0,0,0.1);">

<h3 style="color:red;

font-size:24px;

font-weight:bold;

margin-bottom:15px;">

Weight Training

</h3>

<p>

Build strength and muscle with guided weight training programs.

</p>

</div>

<!-- Card 2 -->

<div style="background-color:white;

padding:25px;

border-radius:10px;

text-align:center;

box-shadow:0 0 10px rgba(0,0,0,0.1);">

<h3 style="color:red;

font-size:24px;

font-weight:bold;

margin-bottom:15px;">

Yoga Classes

</h3>

<p>

Improve flexibility and mental wellness through yoga sessions.

</p>

</div>

<!-- Card 3 -->

<div style="background-color:white;

padding:25px;

border-radius:10px;

text-align:center;

box-shadow:0 0 10px rgba(0,0,0,0.1);">

<h3 style="color:red;

font-size:24px;

font-weight:bold;

margin-bottom:15px;">

Diet Plans

</h3>

<p>

Personalized diet plans to complement your fitness routine.

</p>

</div>

</div>

</section>

<!-- Footer -->

<footer style="background:linear-gradient(to right,#bfdbfe,#818cf8);

color:white;

text-align:center;

padding:20px;">

<p>© 2026 FitLife | All Rights Reserved</p>

<p style="margin-top:8px;">

Contact: info@fitlife.com

</p>

</footer>

</body>

</html>

EXPERIMENT-4

.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Result Calculator</title>
  <link rel="stylesheet" href="Experiment4.css">
</head>
<body>
  <div class="container">
    <h2>Student Result Calculator</h2>
    <label>Enter Number of Subjects</label>
    <input type="number" id="subjects" placeholder="Enter Number of
Subjects"><br><br>
    <button onclick="calculateResult()">Calculate Result</button>
    <div id="result"></div>
  </div>
<script src="main.js"></script>

</body>
</html>
```

.CSS

```
body{
font-family: Arial;
background-color:#f2f2f2;
text-align:center;
padding:40px;
}
.container{
background:white;
padding:25px;
width:400px;
margin:auto;
border-radius:10px;
box-shadow:0px 0px 10px gray;
}
input{
padding:8px;
width:80%;
margin-top:5px;
}
button{
padding:10px 20px;
background-color:blue;
color:white;
border:none;
border-radius:5px;
cursor:pointer;
}
```

```
button:hover{
background-color:darkblue;
}
#result{
margin-top:20px;
font-weight:bold;
font-size:18px;
}
```

.JS

```
function calculateResult(){
let n = document.getElementById("subjects").value;
let total = 0;
for(let i=1;i<=n;i++){
let marks = parseFloat(prompt("Enter marks for Subject " + i));
total = total + marks;
}
let average = total / n;
let grade;
let result;
if(average >= 90){
grade = "A+";
}
else if(average >= 75){
grade = "A";
}
```



```
else if(average >= 60){
    grade = "B";
}
else if(average >= 50){
    grade = "C";
}
else{
    grade = "F";
}
if(average >= 40){
    result = "PASS";
}
else{
    result = "FAIL";
}
document.getElementById("result").innerHTML =
    "Total Marks: " + total + "<br>" +
    "Average Marks: " + average.toFixed(2) + "<br>" +
    "Grade: " + grade + "<br>" +
    "Result: " + result;
}
```

EXPERIMENT-5

.HTML

```
<html>

  <head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Employee Salary Management System</title>

    <link rel="stylesheet" href="Experiment5.css">

  </head>

<body>

  <div class="container">

    <h2>Employee Salary Management System</h2>

    <input type="text" id="name" placeholder="Employee Name"><br>
    <input type="text" id="empld" placeholder="Employee ID"><br>
    <input type="number" id="salary" placeholder="Salary"><br>
    <input type="text" id="dept" placeholder="Department"><br>

    <button onclick="addEmployee()">Add Employee</button><br>
    <button onclick="displayEmployees()">Display All</button>
    <button onclick="filterSalary()">Salary > 50000</button>
    <button onclick="totalSalary()">Total Salary</button>
    <button onclick="averageSalary()">Average Salary</button><br>
    <button onclick="countDepartment()">Count by Department</button>
```

```
<div id="output"></div>
</div>

<script src="main2.js"></script>
</body>
</html>
```

.CSS

```
body {
    font-family: Arial;
    background-color: #eef2f3;
    text-align: center;
}
.container {
    width: 650px;
    margin: 40px auto;
    padding: 20px;
    background: white;
    border-radius: 10px;
    box-shadow: 0 0 10px gray;
}
input {
    padding: 8px;
    margin: 5px;
}
button {
```

```
padding: 8px 12px;
margin: 5px;
background-color: #28a745;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
}
button:hover {
    background-color: #1e7e34;
}
#output {
    margin-top: 20px;
    text-align: left;
}
.JS
```

```
let employees = [];
```

```
function addEmployee() {
    let name = document.getElementById("name").value;
    let empld = document.getElementById("empld").value;
    let salary = parseFloat(document.getElementById("salary").value);
    let dept = document.getElementById("dept").value;

    if (name === "" || empld === "" || isNaN(salary) || dept === "") {
        alert("Please fill all fields properly");
        return;
    }
}
```

```
}
```

```
let employee = {  
  name: name,  
  id: empld,  
  salary: salary,  
  department: dept  
};
```

```
employees.push(employee);  
alert("Employee Added Successfully!");
```

```
document.getElementById("name").value = "";  
document.getElementById("empld").value = "";  
document.getElementById("salary").value = "";  
document.getElementById("dept").value = "";  
}
```

```
function displayEmployees() {  
  let output = "<h3>All Employees</h3>";  
  
  employees.forEach(emp => {  
    output += `  
    Name: ${emp.name} |  
    ID: ${emp.id} |  
    Salary: ₹${emp.salary} |  
    Dept: ${emp.department} <br>  
  `;  
  });  
}
```

```
});
```

```
document.getElementById("output").innerHTML = output;  
}
```

```
function filterSalary() {  
  let filtered = employees.filter(emp => emp.salary > 50000);  
  
  let output = "<h3>Employees with Salary > ₹50,000</h3>";  
  
  filtered.forEach(emp => {  
    output += `  
      Name: ${emp.name} |  
      Salary: ₹${emp.salary} <br>  
    `;  
  });  
});
```

```
document.getElementById("output").innerHTML = output;  
}
```

```
function totalSalary() {  
  let total = employees.reduce((sum, emp) => sum + emp.salary, 0);  
  
  document.getElementById("output").innerHTML =  
    "<h3>Total Salary Payout: ₹" + total + "</h3>";  
}
```

```
function averageSalary() {
```

```
if (employees.length === 0) {  
    document.getElementById("output").innerHTML =  
        "<h3>No employee records available</h3>";  
    return;  
}
```

```
let total = employees.reduce((sum, emp) => sum + emp.salary, 0);  
let avg = total / employees.length;
```

```
document.getElementById("output").innerHTML =  
    "<h3>Average Salary: ₹" + avg.toFixed(2) + "</h3>";  
}
```

```
function countDepartment() {  
    let deptName = prompt("Enter Department Name:");  
    let count = employees.filter(emp => emp.department === deptName).length;  
  
    document.getElementById("output").innerHTML =  
        "<h3>Employees in " + deptName + ": " + count + "</h3>";  
}
```

EXPERIMENT-6

.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>DOM events</title>
  <link rel="stylesheet" href="Experiment6.css">
</head>
<body>

<div class="container">
  <h2 id="mainHeading">Welcome to JavaScript Lab</h2>

  <input type="text" id="userInput" placeholder="Enter new heading text">

  <br><br>

  <button id="changeTextBtn">Change Heading</button>
  <button id="bgColorBtn">Change Background</button>
  <button id="fontSizeBtn">Increase Font Size</button>
  <button id="toggleBtn">Show/Hide Paragraph</button>
  <button id="resetBtn">Reset</button>

  <p id="paragraph">
```


This is a sample paragraph. JavaScript events can dynamically change content.

</p>

</div>

<script src="main3.js"></script>

</body>

</html>

.CSS

body {

font-family: Arial;

background-color: #f4f4f4;

text-align: center;

}

.container {

width: 600px;

margin: 50px auto;

padding: 20px;

background: white;

border-radius: 10px;

box-shadow: 0 0 10px gray;

}

button {

padding: 8px 12px;

```
margin: 5px;
background-color: #007bff;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
}
```

```
button:hover {
  background-color: #0056b3;
}
```

```
input {
  padding: 8px;
  width: 250px;
}
```

.JS

```
let heading = document.getElementById("mainHeading");
let paragraph = document.getElementById("paragraph");
let input = document.getElementById("userInput");
```

```
let fontSize = 16;
```

```
document.getElementById("changeTextBtn").addEventListener("click", function ()
{
  if (input.value !== "") {
```

```
        heading.innerHTML = input.value;
    }
});
```

```
document.getElementById("bgColorBtn").onclick = function () {
    document.body.style.backgroundColor =
        "#" + Math.floor(Math.random() * 16777215).toString(16);
};
```

```
document.getElementById("fontSizeBtn").addEventListener("click", function () {
    fontSize += 2;
    paragraph.style.fontSize = fontSize + "px";
});
```

```
document.getElementById("toggleBtn").addEventListener("click", function () {
    if (paragraph.style.display === "none") {
        paragraph.style.display = "block";
    } else {
        paragraph.style.display = "none";
    }
});
```

```
document.getElementById("resetBtn").addEventListener("click", function () {
    heading.innerHTML = "Welcome to JavaScript Lab";
    paragraph.style.display = "block";
    paragraph.style.fontSize = "16px";
    document.body.style.backgroundColor = "#f4f4f4";
    input.value = "";
});
```

```
fontSize = 16;  
});
```

EXPERIMENT-7

GREETINGS.JSX

```
function Greetings(props) {  
  return (  
    <div className="card">  
      <h2>{props.name}</h2>  
      <p>  
        Course: {props.course}  
      </p>  
      <p>  
        Marks: {props.marks}  
      </p>  
    </div>  
  );  
}  
export default Greetings;
```

APP.JSX

```
import Greetings from "../components/Greetings";  
import "../App.css";  
function App() {  
  return (  
    <div className="container">  
      <h1>Student Information</h1>  
      <div className="cards-container">  
        <Greetings
```

```
        name="Rahul Sharma"
        course="Computer Science"
        marks="85"
    />
    <Greetings
        name="Anita Verma"
        course="Information Technology"
        marks="92"
    />
    <Greetings
        name="Amit Kumar"
        course="Electronics"
        marks="78"
    />
</div>
</div>
);
}
export default App;
```

APP.CSS

```
.container{
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
}
```

```
.heading{
  align-items: center;
  text-align: center;
  font-size: 30px;
  font-weight: bold;
  margin-bottom: 20px;
}

.cards{
  text-align: center;
  width: 300px;
  display: flex;
  flex-direction: column;
  gap: 20px;
}

.card{
  background-color: aliceblue;
  padding : 15px;
  border-radius: 10px;
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
}
```

EXPERIMENT-8

GREETINGS.JSX

```
function Greetings({ count }) {  
  return (  
    <>  
      <h1>React Counter Application</h1>  
      <h2 className="count">{count}</h2>  
    </>  
  );  
}  
export default Greetings;
```

APP.JSX

```
import { useState } from "react";  
import Greetings from "../Components/Greetings";  
import "../App.css";  
const App = () => {  
  const [count, setCount] = useState(0);  
  const increment = () => {  
    setCount(count + 1);  
  };  
  const decrement = () => {  
    setCount(count - 1);  
  };  
};
```



```
const reset = () => {
  setCount(0);
};
return (
  <div className="outer-box">
    <div className="container">
      <Greetings count={count} />
      <div className="buttons">
        <button onClick={increment}>Increment (+)</button>
        <button onClick={decrement}>Decrement (-)</button>
      </div>
      <button className="reset-btn" onClick={reset}>
        Reset
      </button>
    </div>
  </div>
);
}
export default App;
```

APP.CSS

```
body {
  margin: 0;
  font-family: 'Segoe UI', Arial, sans-serif;
  background-color: #e9ecef;
}
.outer-box {
```

```
height: 100vh;
display: flex;
justify-content: center;
align-items: center;
}
.container {
background: #ffffff;
padding: 45px 60px;
border-radius: 16px;
box-shadow: 0 10px 25px rgba(0, 0, 0, 0.15);
text-align: center;
width: 420px;
transition: 0.3s ease;
}
h1 {
font-size: 28px;
color: #1f2937;
margin-bottom: 15px;
font-weight: 600;
}
.count {
font-size: 52px;
font-weight: bold;
color: #111827;
margin: 20px 0 30px;
}
.buttons {
display: flex;
```

```
justify-content: center;
gap: 15px;
margin-bottom: 20px;
}
button {
background: linear-gradient(135deg, #1e73d8, #2563eb);
color: white;
padding: 10px 20px;
border-radius: 8px;
cursor: pointer;
font-size: 14px;
font-weight: 500;
transition: all 0.25s ease;
}
.reset-btn {
display: block;
margin: 10px auto 0;
padding: 10px 25px;
}
```

EXPERIMENT-9

APP.JSX

```
import React, { useState } from "react";
import "./App.css";
const App = () => {
  const [name, setName] = useState("");
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const [errors, setErrors] = useState({});
  const [users, setUsers] = useState([]);
  const [success, setSuccess] = useState("");
  const validateForm = () => {
    let newErrors = {};
    if (name.trim() === "") {
      newErrors.name = "Name is required";
    }
    if (email.trim() === "") {
      newErrors.email = "Email is required";
    }
    if (password.length < 6) {
      newErrors.password =
        "Password must be at least 6 characters";
    }
    return newErrors;
  };
  const handleSubmit = (e) => {
```

```

e.preventDefault();

const validationErrors = validateForm();

if (Object.keys(validationErrors).length > 0) {
  setErrors(validationErrors);
  setSuccess("");
} else {
  setErrors({});

  const newUser = {
    id: Date.now(),
    name,
    email,
  };
  setUsers([...users, newUser]);
  setSuccess("Registration Successful!");
  setName("");
  setEmail("");
  setPassword("");
}
};

return (
  <div className="main-container">
    <div className="form-container">
      <h1>Registration Form</h1>
      <form onSubmit={handleSubmit}>
        <input
          type="text"
          placeholder="Enter Name"

```

```
        value={name}
        onChange={(e) => setName(e.target.value)}
    />
    {errors.name && (
        <p className="error">{errors.name}</p>
    )}
    <input
        type="email"
        placeholder="Enter Email"
        value={email}
        onChange={(e) => setEmail(e.target.value)}
    />
    {errors.email && (
        <p className="error">{errors.email}</p>
    )}
    <input
        type="password"
        placeholder="Enter Password"
        value={password}
        onChange={(e) => setPassword(e.target.value)}
    />
    {errors.password && (
        <p className="error">{errors.password}</p>
    )}
    <button type="submit">Register</button>
</form>
{success && (
    <h3 className="success">{success}</h3>
```

```

    })
    {users.length > 0 && (
      <div className="users-box">
        <h2>Registered Users</h2>
        <ul>
          {users.map((user) => (
            <li key={user.id}>
              {user.name} - {user.email}
            </li>
          ))}
        </ul>
      </div>
    )}
  </div>
</div>
);
}
export default App;

```

APP.CSS

```

* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}
body {
  background-color: #e6e6e6;

```

```
    font-family: Arial, sans-serif;
}

.main-container {
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
    padding: 20px;
}

.form-container {
    width: 350px;
    background-color: #f3f3f3;
    padding: 35px;
    border-radius: 10px;
    box-shadow: 0px 0px 10px rgba(0, 0, 0, 0.2);
}

.form-container h1 {
    text-align: center;
    margin-bottom: 35px;
    color: #111;
    font-size: 34px;
}

form {
    display: flex;
    flex-direction: column;
}

form input {
    padding: 12px;
```



```
margin-bottom: 20px;
border: 1px solid #999;
font-size: 15px;
outline: none;
}
form input:focus {
    border: 1px solid green;
}
button {
    width: 110px;
    margin: auto;
    padding: 10px;
    background-color: green;
    color: white;
    border: none;
    cursor: pointer;
    font-size: 15px;
}
button:hover {
    background-color: darkgreen;
}
.error {
    color: red;
    font-size: 13px;
    margin-top: -15px;
    margin-bottom: 15px;
}
```

```
.success {  
    text-align: center;  
    color: green;  
    margin-top: 20px;  
    font-size: 20px;  
}  
.users-box {  
    margin-top: 25px;  
    background-color: #dfe9dc;  
    padding: 20px;  
}  
.users-box h2 {  
    text-align: center;  
    margin-bottom: 15px;  
}  
.users-box ul {  
    padding-left: 20px;  
}  
.users-box li {  
    margin-bottom: 10px;  
    font-size: 15px;  
}
```

EXPERIMENT-10

.JAVA

```
abstract class BankAccount {
    int accountNumber;
    String accountHolderName;
    double balance;

    BankAccount(int accountNumber, String accountHolderName, double balance)
    {
        this.accountNumber = accountNumber;
        this.accountHolderName = accountHolderName;
        this.balance = balance;
    }

    void deposit(double amount) {
        balance = balance + amount;
        System.out.println("Amount Deposited: " + amount);
    }

    void displayDetails() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Account Holder Name: " + accountHolderName);
        System.out.println("Balance: " + balance);
    }

    abstract void calculateInterest();
}
```

```
}
```

```
class SavingsAccount extends BankAccount {
```

```
    SavingsAccount(int accountNumber, String accountHolderName, double  
balance) {
```

```
        super(accountNumber, accountHolderName, balance);
```

```
    }
```

```
    void calculateInterest() {
```

```
        double interest = balance * 0.05;
```

```
        System.out.println("Savings Account Interest: " + interest);
```

```
    }
```

```
}
```

```
class CurrentAccount extends BankAccount {
```

```
    CurrentAccount(int accountNumber, String accountHolderName, double  
balance) {
```

```
        super(accountNumber, accountHolderName, balance);
```

```
    }
```

```
    void calculateInterest() {
```

```
        double interest = balance * 0.02;
```

```
        System.out.println("Current Account Interest: " + interest);
```

```
    }
```

```
}
```

```
public class Exp10 {
```

```
    public static void main(String[] args) {
```

```
SavingsAccount sa = new SavingsAccount(101, "Rahul", 10000);
CurrentAccount ca = new CurrentAccount(102, "Anita", 20000);
System.out.println("----- Savings Account -----");
sa.deposit(2000);
sa.displayDetails();
sa.calculateInterest();
System.out.println();
System.out.println("----- Current Account -----");
ca.deposit(3000);
ca.displayDetails();
ca.calculateInterest();
}
}
```

EXPERIMENT-11

SAMBHAV.JAVA

```
package com.abes.JavaSambhav;

public class Sambhav {

    private int id;
    private String name;
    private String course;

    public Sambhav() {
    }

    public Sambhav(int id, String name, String course) {
        this.id = id;
        this.name = name;
        this.course = course;
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }
}
```

```
public String getName() {  
    return name;  
}  
  
public void setName(String name) {  
    this.name = name;  
}  
  
public String getCourse() {  
    return course;  
}  
  
public void setCourse(String course) {  
    this.course = course;  
}  
}
```

SAMBHAVCONTROLLER.JAVA

```
package com.abes.JavaSambhav;  
  
import org.springframework.http.HttpStatus;  
import org.springframework.http.ResponseEntity;  
import org.springframework.web.bind.annotation.*;  
  
@RestController
```

```

@RequestMapping("/")
public class SambhavController {

    private Sambhav student;

    @GetMapping
    public String welcomeMessage() {
        return "Welcome to Management REST API";
    }

    @GetMapping("/student")
    public ResponseEntity<Sambhav> getStudent() {
        if (student == null) {
            return new ResponseEntity<>(HttpStatus.NOT_FOUND);
        }
        return new ResponseEntity<>(student, HttpStatus.OK);
    }

    @PostMapping("/student")
    public ResponseEntity<Sambhav> createStudent(@RequestBody Sambhav
student) {

        this.student = student;
        return new ResponseEntity<>(student, HttpStatus.CREATED);
    }
}

```

SAMBHAVRESTAPPLICATION.JAVA

```
package com.abes.JavaSambhav;
```



```
import org.springframework.boot.SpringApplication;  
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
@SpringBootApplication
```

```
public class SambhavRestApplication {
```

```
    public static void main(String[] args) {
```

```
        SpringApplication.run(SambhavRestApplication.class, args);
```

```
    }
```

```
}
```

EXPERIMENT-12

EMPLOYEE.JAVA

```
package com.abes.EmployeeAPI.model;

import jakarta.persistence.Entity;
import jakarta.persistence.Id;

public class Employee {

    private int id;
    private String name;
    private String department;
    private double salary;

    public Employee() {}

    public Employee(int id, String name, String department, double salary) {
        this.id = id;
        this.name = name;
        this.department = department;
        this.salary = salary;
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
```

```
        this.id = id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getDepartment() {
        return department;
    }

    public void setDepartment(String department) {
        this.department = department;
    }

    public double getSalary() {
        return salary;
    }

    public void setSalary(double salary) {
        this.salary = salary;
    }
}
```

EMPLOYEECONTROLLER.JAVA

```
package com.abes.EmployeeAPI.controller;
```

```
import java.util.List;
```

```
import org.springframework.beans.factory.annotation.Autowired;
```

```
import org.springframework.web.bind.annotation.*;
```

```
import com.abes.EmployeeAPI.model.Employee;
```

```
import com.abes.EmployeeAPI.service.EmployeeService;
```

```
public class EmployeeController {
```

```
    private EmployeeService service;
```

```
    public Employee addEmployee(@RequestBody Employee employee) {
```

```
        return service.saveEmployee(employee);
```

```
    }
```

```
    public List<Employee> getAllEmployees() {
```

```
        return service.getAllEmployees();
```

```
    }
```

```
    @GetMapping("/{id}")
```

```
    public Employee getEmployee(@PathVariable int id) {
```

```
        return service.getEmployeeById(id);
```

```
    }
```

```
    @DeleteMapping("/{id}")
```

```
    public void deleteEmployee(@PathVariable int id) {
```

```
        service.deleteEmployee(id);
```

```
    }
```

```
}
```

EMPLOYEE REPOSITORY.JAVA

```
package com.abes.EmployeeAPI.repository;
```

```
import org.springframework.data.jpa.repository.JpaRepository;
```

```
import com.abes.EmployeeAPI.model.Employee;
```

```
public interface EmployeeRepository extends JpaRepository<Employee, Integer>
{
}
}
```

EMPLOYEE SERVICE.JAVA

```
package com.abes.EmployeeAPI.service;
```

```
import java.util.List;
```

```
import org.springframework.beans.factory.annotation.Autowired;
```

```
import org.springframework.stereotype.Service;
```

```
import com.abes.EmployeeAPI.model.Employee;
```

```
import com.abes.EmployeeAPI.repository.EmployeeRepository;
```

```
public class EmployeeService {
```

```
    private EmployeeRepository repository;
```

```
    public Employee saveEmployee(Employee employee) {
        return repository.save(employee);
    }
}
```

```

    }

    public List<Employee> getAllEmployees() {
        return repository.findAll();
    }

    public Employee getEmployeeById(int id) {
        return repository.findById(id).orElse(null);
    }

    public void deleteEmployee(int id) {
        repository.deleteById(id);
    }
}

```

EMPLOYEEAPIAPPLICATION.JAVA

```

package com.abes.EmployeeAPI;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication

public class EmployeeApiApplication {

    public static void main(String[] args) {
        SpringApplication.run(EmployeeApiApplication.class, args);
    }
}

```

}